Quiz

Mar. 23, 2026

多选题 2分

Which of the following is/are true about the base & bounds scheme?

A It requires a pair of hardware registers (base and bounds) per CPU to perform address translation and bounds checking.

B It effectively eliminates all forms of memory fragmentation.

C It makes inter-process memory sharing, such as sharing code segments, straightforward and easy to implement.

D It can lead to internal fragmentation, where memory between the heap and stack within a process's allocated slot is wasted.

E The operating system must manage physical memory allocation by carving out contiguous slots for each process's address space.

提交

多选题 2分

Which of the followings is/are true about the segmentation scheme?

A. Segmentation allows each segment (e.g., code, heap, stack) to be placed independently in different regions of physical memory.

B. Segmentation completely eliminates external fragmentation, making physical memory allocation trivial for the operating system.

C. Segmentation enables memory sharing between processes by allowing multiple processes to map their segments to the same physical memory region (e.g., for shared libraries).

D. Segmentation can be implemented with base/bounds register pairs organized as a table, indexed by a Segment ID derived from the virtual address.

E. Managing free spaces in physical memory with variable-sized segments can lead to external fragmentation, where free gaps exist between allocated segments.

提交

多选题 2分

Which of the following is/are true about memory allocation strategies?

A First Fit is the slowest strategy because it must exhaustively search the entire free list to find the smallest suitable block.

B Best Fit aims to minimize external fragmentation by returning the smallest block that satisfies the request, but it suffers from slow exhaustive search.

C Worst Fit is designed to leave larger holes in physical memory by using the largest available block, which helps reduce fragmentation in practice.

D First Fit is fast because it returns the first block that is big enough, but it can pollute the beginning of the free list with small chunks.

E Best Fit and Worst Fit both require exhaustive search of the free list, making them slower than First Fit in practice.

提交

多选题 2分

Which of the following is/are true about paging?

- A Paging divides physical memory into fixed-sized blocks called frames and virtual memory into same-sized blocks called pages, eliminating external fragmentation.
- B Paging requires the entire process address space to be contiguously mapped to a single region of physical memory, similar to Base & Bounds.
- C Paging uses a page table maintained by the operating system to map virtual pages to physical frames, allowing non-contiguous allocation of a process's memory.
- D Paging completely eliminates the need for segmentation by using a single-level page table that requires no hardware support from the CPU.
- E Page to page frame mapping can be one-to-one, one-to-many, and many-to-one.

提交

多选题 2分

Which of the followings is/are true about the design philosophy of page tables?

A Compared to single-level page tables, the design of multi-level page tables aims to reduce the time complexity of address translation, at the cost of increasing space complexity.

B For simplicity of memory management, each level of page tables should fit in one single page.

C Page tables enable isolation and protection by ensuring that each process has its own page table, preventing one process from accessing another process's physical memory.

D If inverted page table is used, each process should have its own inverted page table.

E Page tables are designed to be entirely software-managed without hardware involvement to maximize flexibility and simplify CPU design

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